

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12418850
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Sood; Romil Kumar

Disabling roaming in overlapping coverage areas of wireless telecommunications networks

Abstract

A telecommunications network provides selective roaming for user equipment (UE) devices registered to the network, allowing some UE devices to operate on a roaming network in some geographic regions and disallowing other UE devices to access roaming networks. While a target UE device is operating on a roaming network, the telecommunications network monitors changes in location of the target device. If the target UE device is detected to have moved into an overlapping coverage area, covered by both the telecommunications network and the roaming network, the telecommunications network generates a command to cancel the subscription of the target UE device to the roaming network.

Inventors:	Sood; Romil Kumar (Bothell, WA)
Applicant:	T-Mobile USA, Inc. (Bellevue, WA)
Family ID:	1000008819797
Assignee:	T-Mobile USA, Inc. (Bellevue, WA)
Appl. No.:	17/897097
Filed:	August 26, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240073784 A1	Feb. 29, 2024

Publication Classification

Int. Cl.:	H04W48/04 (20090101); H04W8/02 (20090101)
U.S. Cl.:	

Field of Classification Search

CPC: H04W (48/04); H04W (8/02); H04W (8/12)

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6542497	12/2002	Curry et al.	N/A	N/A
6560223	12/2002	Egan et al.	N/A	N/A
6680922	12/2003	Jorgensen	N/A	N/A
6757268	12/2003	Zendle	N/A	N/A
6862622	12/2004	Jorgensen	N/A	N/A
6865170	12/2004	Zendle	N/A	N/A
7010002	12/2005	Chow et al.	N/A	N/A
7035932	12/2005	Dowling	N/A	N/A
7047008	12/2005	Martlew	N/A	N/A
7133670	12/2005	Moll et al.	N/A	N/A
7133678	12/2005	Berkowitz et al.	N/A	N/A
7142843	12/2005	Dowling et al.	N/A	N/A
7146636	12/2005	Crosbie	N/A	N/A
7151931	12/2005	Tsao et al.	N/A	N/A
7171209	12/2006	Numminen et al.	N/A	N/A
7197569	12/2006	Dowling	N/A	N/A
7292592	12/2006	Rune	N/A	N/A
7293110	12/2006	Dowling	N/A	N/A
7299069	12/2006	Claussen	N/A	N/A
7305229	12/2006	Fox et al.	N/A	N/A
7433929	12/2007	Guilford et al.	N/A	N/A
7483698	12/2008	Hurtta	N/A	N/A
7581030	12/2008	Dowling	N/A	N/A
7613142	12/2008	Molteni et al.	N/A	N/A
7634269	12/2008	Gallagher	N/A	N/A
7640008	12/2008	Gallagher et al.	N/A	N/A
7747251	12/2009	Pippert et al.	N/A	N/A
7805161	12/2009	Jones et al.	N/A	N/A
7843900	12/2009	Gallagher et al.	N/A	N/A
7894812	12/2010	Durig et al.	N/A	N/A
7899451	12/2010	Hu et al.	N/A	N/A
7933595	12/2010	Pippert et al.	N/A	N/A
7949326	12/2010	Gallagher et al.	N/A	N/A
7979086	12/2010	Jones et al.	N/A	N/A
8045493	12/2010	Gallagher et al.	N/A	N/A
8060084	12/2010	Buckley et al.	N/A	N/A
8078164	12/2010	Ganesan	N/A	N/A
8086239	12/2010	Elmaleh	N/A	N/A
8144672	12/2011	Pandey et al.	N/A	N/A

8189548	12/2011	Caldwell et al.	N/A	N/A
8204543	12/2011	Keevill et al.	N/A	N/A
8213424	12/2011	Jabbari et al.	N/A	N/A
8228870	12/2011	Sigg et al.	N/A	N/A
8233899	12/2011	Oroskar et al.	N/A	N/A
8238906	12/2011	Oroskar et al.	N/A	N/A
8259666	12/2011	Bao et al.	N/A	N/A
8279800	12/2011	Schmidt et al.	N/A	N/A
8311536	12/2011	Pulugurta	N/A	N/A
RE43856	12/2011	Berkowitz et al.	N/A	N/A
8363626	12/2012	Faccin	N/A	N/A
8391858	12/2012	Vargantwar et al.	N/A	N/A
8391859	12/2012	Pulugurta	N/A	N/A
8406757	12/2012	Singh et al.	N/A	N/A
8472946	12/2012	Chan et al.	N/A	N/A
8483183	12/2012	Liu et al.	N/A	N/A
8494522	12/2012	Chen et al.	N/A	N/A
8510802	12/2012	Singh et al.	N/A	N/A
8625537	12/2013	Sigg et al.	N/A	N/A
8644818	12/2013	Okmyanskiy et al.	N/A	N/A
8730796	12/2013	Samuel et al.	N/A	N/A
8886201	12/2013	Fang et al.	N/A	N/A
9137735	12/2014	Thomas et al.	N/A	N/A
9167410	12/2014	Xue et al.	N/A	N/A
9325621	12/2015	Ramamurthy et al.	N/A	N/A
9432924	12/2015	Singh et al.	N/A	N/A
9596707	12/2016	Zhu et al.	N/A	N/A
9622192	12/2016	Ji	N/A	N/A
9736700	12/2016	Douberley et al.	N/A	N/A
9749899	12/2016	Raleigh et al.	N/A	N/A
9781588	12/2016	Cho et al.	N/A	N/A
9832678	12/2016	Tandon et al.	N/A	N/A
9832797	12/2016	Wang et al.	N/A	N/A
9860827	12/2017	Ji et al.	N/A	N/A
9877253	12/2017	Liu et al.	N/A	N/A
9900765	12/2017	Gonzalez et al.	N/A	N/A
9913154	12/2017	Grischy et al.	N/A	N/A
9923657	12/2017	Forenza et al.	N/A	N/A
9949166	12/2017	Sirotkin et al.	N/A	N/A
9974091	12/2017	Rubin et al.	N/A	N/A
10271298	12/2018	Yang	N/A	H04W 48/16
10735940	12/2019	Sharma	N/A	N/A
10984128	12/2020	Hoffer	N/A	N/A
11240729	12/2021	Xu	N/A	H04L 45/123
2001/0055298	12/2000	Baker et al.	N/A	N/A
2002/0087674	12/2001	Guilford et al.	N/A	N/A
2002/0136226	12/2001	Christoffel et al.	N/A	N/A
2002/0147008	12/2001	Kallio	N/A	N/A
2002/0181444	12/2001	Acampora	N/A	N/A
2003/0036374	12/2002	English et al.	N/A	N/A

2005/0009525	12/2004	Evslin	N/A	N/A
2005/0030907	12/2004	Lou et al.	N/A	N/A
2005/0088999	12/2004	Waylett et al.	N/A	N/A
2005/0232193	12/2004	Jorgensen	N/A	N/A
2006/0098593	12/2005	Edvardsen et al.	N/A	N/A
2006/0135160	12/2005	Jiang	N/A	N/A
2006/0253526	12/2005	Welch et al.	N/A	N/A
2007/0121540	12/2006	Sharp et al.	N/A	N/A
2007/0160017	12/2006	Meier et al.	N/A	N/A
2007/0201469	12/2006	Iyer et al.	N/A	N/A
2008/0089287	12/2007	Sagfors et al.	N/A	N/A
2009/0086672	12/2008	Gholmieh et al.	N/A	N/A
2010/0048205	12/2009	Guilford et al.	N/A	N/A
2010/0075669	12/2009	Sparks et al.	N/A	N/A
2010/0093344	12/2009	Chan et al.	N/A	N/A
2010/0184428	12/2009	Luo	N/A	N/A
2010/0234022	12/2009	Winterbottom	N/A	N/A
2013/0288671	12/2012	Keller	455/433	H04W 8/02
2014/0169286	12/2013	Xu	370/329	H04W 8/02
2014/0293982	12/2013	Gupta	370/338	H04L 41/08
2017/0135036	12/2016	Schmidt et al.	N/A	N/A
2019/0289571	12/2018	Park	N/A	H04W 60/00
2019/0327663	12/2018	Wirth et al.	N/A	N/A
2020/0128449	12/2019	Faus Gregori	N/A	H04W 36/00222
2021/0136658	12/2020	Rönneke	N/A	H04W 48/06
2021/0168695	12/2020	Ryu	N/A	H04W 8/14
2022/0124480	12/2021	Huang	N/A	H04W 8/12
2022/0159445	12/2021	Rajavelu	N/A	H04W 8/12
2022/0322066	12/2021	Sood et al.	N/A	N/A
2022/0330004	12/2021	Sood et al.	N/A	N/A
2022/0417758	12/2021	Nooren	N/A	H04L 47/80
2023/0106631	12/2022	Venkataraman et al.	N/A	N/A
2023/0276199	12/2022	Ryu	455/456.1	H04W 84/18
2023/0291609	12/2022	Lonkar	N/A	N/A
2023/0308852	12/2022	Srivastava et al.	N/A	N/A
2024/0064503	12/2023	Sihotang et al.	N/A	N/A
2024/0064599	12/2023	Sood et al.	N/A	N/A
2024/0073671	12/2023	Sood	N/A	N/A
2024/0073784	12/2023	Sood	N/A	N/A
2024/0205663	12/2023	Arends et al.	N/A	N/A
2024/0259788	12/2023	Khurana	N/A	N/A
2024/0267734	12/2023	Roemer	N/A	H04W 12/088
2024/0275760	12/2023	Kaur et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102244857	12/2014	CN	N/A
104661280	12/2014	CN	N/A
104378751	12/2018	CN	N/A

113873522	12/2020	CN	N/A
109196890	12/2021	CN	N/A
102017125042	12/2020	DE	N/A
1665640	12/2005	EP	N/A
2391157	12/2014	EP	N/A
2621834	12/2023	GB	N/A
2621835	12/2023	GB	N/A
2009207164	12/2008	JP	N/A
4852339	12/2010	JP	N/A
6372726	12/2017	JP	N/A
6885970	12/2020	JP	N/A
100432462	12/2003	KR	N/A
101073372	12/2010	KR	N/A
101486740	12/2014	KR	N/A
20160108472	12/2015	KR	N/A
102288207	12/2020	KR	N/A
102346610	12/2021	KR	N/A
9858477	12/1997	WO	N/A
9904534	12/1998	WO	N/A
03084096	12/2002	WO	N/A
2005004523	12/2004	WO	N/A
2005025138	12/2004	WO	N/A
2005039115	12/2004	WO	N/A
2006120533	12/2005	WO	N/A
2012019317	12/2011	WO	N/A
2013019551	12/2012	WO	N/A
2014051630	12/2013	WO	N/A
2014084910	12/2013	WO	N/A
2014149641	12/2013	WO	N/A
2014185987	12/2013	WO	N/A
2015199881	12/2014	WO	N/A
2017059891	12/2016	WO	N/A
2017210447	12/2016	WO	N/A
2018127505	12/2017	WO	N/A
2022026143	12/2021	WO	N/A
2024149512	12/2023	WO	N/A

Primary Examiner: Rahman; Shah M

Attorney, Agent or Firm: Perkins Coie LLP

Background/Summary

BACKGROUND

(1) Wireless telecommunications networks typically operate in limited geographic regions. Under some circumstances, wireless devices registered to one operator's network desire to operate outside of the geographic region supported by that network. If another telecommunications network operates in the region, the wireless device can use the other network as a roaming network.

However, the infrastructure to support roaming to other telecommunications networks can be expensive and burdensome for a telecommunications network to maintain.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Detailed descriptions of implementations of the present invention will be described and explained through the use of the accompanying drawings.
- (2) FIG. 1 is a block diagram that illustrates a wireless communications system that can implement aspects of the present technology.
- (3) FIG. 2 is a block diagram that illustrates 5G core network functions (NFs) that can implement aspects of the present technology.
- (4) FIG. 3 is a block diagram that illustrates 4G core functions that can implement aspects of the present technology.
- (5) FIG. 4 illustrates an example network environment in which selective roaming is enabled.
- (6) FIG. 5 illustrates a selective roaming process performed between 4G networks, according to some implementations.
- (7) FIG. 6 illustrates a selective roaming process performed between 5G networks, according to some implementations.
- (8) FIG. 7 illustrates an example overlapping coverage area between a home public land mobile network and a visiting public land mobile network.
- (9) FIG. 8 illustrates a first example process to disconnect a user equipment device from a roaming network, according to some implementations.
- (10) FIG. 9 illustrates a second example process to disconnect a user equipment device from a roaming network, according to some implementations.
- (11) FIG. 10 illustrates a third example process to disconnect a user equipment device from a roaming network, according to some implementations.
- (12) FIG. 11 is a block diagram that illustrates an example of a computer system in which at least some operations described herein can be implemented.
- (13) The technologies described herein will become more apparent to those skilled in the art from studying the Detailed Description in conjunction with the drawings. Embodiments or implementations describing aspects of the invention are illustrated by way of example, and the same references can indicate similar elements. While the drawings depict various implementations for the purpose of illustration, those skilled in the art will recognize that alternative implementations can be employed without departing from the principles of the present technologies. Accordingly, while specific implementations are shown in the drawings, the technology is amenable to various modifications.

DETAILED DESCRIPTION

- (14) Some customers of telecommunications networks desire to enable roaming functionality that allows the customer to occasionally use service of another telecommunications network. For example, a customer who lives or works in an area with limited coverage by their registered home cellular/telecommunications network, or who frequently travels to an area with limited or no coverage by the home network, may desire to use service of another telecommunications network that has more reliable coverage in those areas. Under conventional techniques, a telecommunications network typically must either enable roaming for all its customers or for none of its customers. Because roaming support can be expensive and burdensome for a telecommunications network to maintain, an operator of the network may not want to enable roaming functionality for all its customers.
- (15) To support a subset of customers for whom roaming functionality improves the customer's

experience with a telecommunications network while limiting the burden of supporting roaming infrastructure, the inventors have conceived of and reduced to practice techniques to enable selective roaming in a wireless telecommunications network. According to implementations described herein, the telecommunications network maintains a set of roaming permissions that specify whether a particular wireless device is permitted to access roaming networks, as well as particular geographic region(s) in which the device can access a roaming network. When a wireless device requests to attach to a roaming network, the telecommunications network queries the roaming permissions and grants the request if the wireless device is accessing a permitted roaming network in a permitted location.

(16) When a wireless device, while attached to a roaming network, moves into an area of overlapping coverage between the roaming network and the device's home network, the home network may desire to cause the wireless device to operate on the home network rather than the roaming network because access to the roaming network is not needed in the overlapping coverage area. To identify when a wireless device moves into the overlapping coverage area, the home network monitors changes in location of the wireless device by, for example, receiving Tracking Area Updates (TAUs) generated by the wireless device or receiving Modify Bearer Requests (MBRs) generated by the wireless device. The location of the wireless device is decoded from the TAU or MBR and compared to known coverage areas of the home network or known overlapping coverage areas between the home and roaming networks. When the wireless device is determined to be located in an overlapping coverage area, the home network generates a command to cancel the subscription of the wireless device to the roaming network.

(17) The description and associated drawings are illustrative examples and are not to be construed as limiting. This disclosure provides certain details for a thorough understanding and enabling description of these examples. One skilled in the relevant technology will understand, however, that the invention can be practiced without many of these details. Likewise, one skilled in the relevant technology will understand that the invention can include well-known structures or features that are not shown or described in detail, to avoid unnecessarily obscuring the descriptions of examples.

(18) Wireless Communications System

(19) FIG. 1 is a block diagram that illustrates a wireless telecommunication network **100** (“network **100**”) in which aspects of the disclosed technology are incorporated. The network **100** includes base stations **102-1** through **102-4** (also referred to individually as “base station **102**” or collectively as “base stations **102**”). A base station is a type of network access node (NAN) that can also be referred to as a cell site, a base transceiver station, or a radio base station. The network **100** can include any combination of NANs including an access point, radio transceiver, gNodeB (gNB), NodeB, eNodeB (eNB), Home NodeB or Home eNodeB, or the like. In addition to being a wireless wide area network (WWAN) base station, a NAN can be a wireless local area network (WLAN) access point, such as an Institute of Electrical and Electronics Engineers (IEEE) 802.11 access point.

(20) The NANs of a network **100** formed by the network **100** also include wireless devices **104-1** through **104-7** (referred to individually as “wireless device **104**” or collectively as “wireless devices **104**”) and a core network **106**. The wireless devices **104-1** through **104-7** can correspond to or include network **100** entities capable of communication using various connectivity standards. For example, a 5G communication channel can use millimeter wave (mmW) access frequencies of 28 GHz or more. In some implementations, the wireless device **104** can operatively couple to a base station **102** over a long-term evolution/long-term evolution-advanced (LTE/LTE-A) communication channel, which is referred to as a 4G communication channel.

(21) The core network **106** provides, manages, and controls security services, user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. The base stations **102** interface with the core network **106** through a first set of

backhaul links (e.g., S1 interfaces) and can perform radio configuration and scheduling for communication with the wireless devices **104** or can operate under the control of a base station controller (not shown). In some examples, the base stations **102** can communicate with each other, either directly or indirectly (e.g., through the core network **106**), over a second set of backhaul links **110-1** through **110-3** (e.g., X1 interfaces), which can be wired or wireless communication links.

(22) The base stations **102** can wirelessly communicate with the wireless devices **104** via one or more base station antennas. The cell sites can provide communication coverage for geographic coverage areas **112-1** through **112-4** (also referred to individually as “coverage area **112**” or collectively as “coverage areas **112**”). The geographic coverage area **112** for a base station **102** can be divided into sectors making up only a portion of the coverage area (not shown). The network **100** can include base stations of different types (e.g., macro and/or small cell base stations). In some implementations, there can be overlapping geographic coverage areas **112** for different service environments (e.g., Internet-of-Things (IoT), mobile broadband (MBB), vehicle-to-everything (V2X), machine-to-machine (M2M), machine-to-everything (M2X), ultra-reliable low-latency communication (URLLC), machine-type communication (MTC), etc.).

(23) The network **100** can include a 5G network **100** and/or an LTE/LTE-A or other network. In an LTE/LTE-A network, the term eNB is used to describe the base stations **102**, and in 5G new radio (NR) networks, the term gNBs is used to describe the base stations **102** that can include mmW communications. The network **100** can thus form a heterogeneous network **100** in which different types of base stations provide coverage for various geographic regions. For example, each base station **102** can provide communication coverage for a macro cell, a small cell, and/or other types of cells. As used herein, the term “cell” can relate to a base station, a carrier or component carrier associated with the base station, or a coverage area (e.g., sector) of a carrier or base station, depending on context.

(24) A macro cell generally covers a relatively large geographic area (e.g., several kilometers in radius) and can allow access by wireless devices that have service subscriptions with a wireless network **100** service provider. As indicated earlier, a small cell is a lower-powered base station, as compared to a macro cell, and can operate in the same or different (e.g., licensed, unlicensed) frequency bands as macro cells. Examples of small cells include pico cells, femto cells, and micro cells. In general, a pico cell can cover a relatively smaller geographic area and can allow unrestricted access by wireless devices that have service subscriptions with the network **100** provider. A femto cell covers a relatively smaller geographic area (e.g., a home) and can provide restricted access by wireless devices having an association with the femto unit (e.g., wireless devices in a closed subscriber group (CSG), wireless devices for users in the home). A base station can support one or multiple (e.g., two, three, four, and the like) cells (e.g., component carriers). All fixed transceivers noted herein that can provide access to the network **100** are NANs, including small cells.

(25) The communication networks that accommodate various disclosed examples can be packet-based networks that operate according to a layered protocol stack. In the user plane, communications at the bearer or Packet Data Convergence Protocol (PDCP) layer can be IP-based. A Radio Link Control (RLC) layer then performs packet segmentation and reassembly to communicate over logical channels. A Medium Access Control (MAC) layer can perform priority handling and multiplexing of logical channels into transport channels. The MAC layer can also use Hybrid ARQ (HARQ) to provide retransmission at the MAC layer, to improve link efficiency. In the control plane, the Radio Resource Control (RRC) protocol layer provides establishment, configuration, and maintenance of an RRC connection between a wireless device **104** and the base stations **102** or core network **106** supporting radio bearers for the user plane data. At the Physical (PHY) layer, the transport channels are mapped to physical channels.

(26) Wireless devices can be integrated with or embedded in other devices. As illustrated, the wireless devices **104** are distributed throughout the wireless telecommunications network **100**,

where each wireless device **104** can be stationary or mobile. For example, wireless devices can include handheld mobile devices **104-1** and **104-2** (e.g., smartphones, portable hotspots, tablets, etc.); laptops **104-3**; wearables **104-4**; drones **104-5**; vehicles with wireless connectivity **104-6**; head-mounted displays with wireless augmented reality/virtual reality (AR/VR) connectivity **104-7**; portable gaming consoles; wireless routers, gateways, modems, and other fixed-wireless access devices; wirelessly connected sensors that provides data to a remote server over a network; IoT devices such as wirelessly connected smart home appliances, etc.

(27) A wireless device (e.g., wireless devices **104-1**, **104-2**, **104-3**, **104-4**, **104-5**, **104-6**, and **104-7**) can be referred to as a user equipment (UE), a customer premise equipment (CPE), a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a handheld mobile device, a remote device, a mobile subscriber station, terminal equipment, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a mobile client, a client, or the like.

(28) A wireless device can communicate with various types of base stations and network **100** equipment at the edge of a network **100** including macro eNBs/gNBs, small cell eNBs/gNBs, relay base stations, and the like. A wireless device can also communicate with other wireless devices either within or outside the same coverage area of a base station via device-to-device (D2D) communications.

(29) The communication links **114-1** through **114-9** (also referred to individually as “communication link **114**” or collectively as “communication links **114**”) shown in network **100** include uplink (UL) transmissions from a wireless device **104** to a base station **102**, and/or downlink (DL) transmissions from a base station **102** to a wireless device **104**. The downlink transmissions can also be called forward link transmissions while the uplink transmissions can also be called reverse link transmissions. Each communication link **114** includes one or more carriers, where each carrier can be a signal composed of multiple sub-carriers (e.g., waveform signals of different frequencies) modulated according to the various radio technologies. Each modulated signal can be sent on a different sub-carrier and carry control information (e.g., reference signals, control channels), overhead information, user data, etc. The communication links **114** can transmit bidirectional communications using frequency division duplex (FDD) (e.g., using paired spectrum resources) or time division duplex (TDD) operation (e.g., using unpaired spectrum resources). In some implementations, the communication links **114** include LTE and/or mmW communication links.

(30) In some implementations of the network **100**, the base stations **102** and/or the wireless devices **104** include multiple antennas for employing antenna diversity schemes to improve communication quality and reliability between base stations **102** and wireless devices **104**. Additionally or alternatively, the base stations **102** and/or the wireless devices **104** can employ multiple-input, multiple-output (MIMO) techniques that can take advantage of multi-path environments to transmit multiple spatial layers carrying the same or different coded data.

(31) 5G Core Network Functions

(32) FIG. 2 is a block diagram that illustrates an architecture **200** including 5G core network functions (NFs) that can implement aspects of the present technology. A wireless device **202** can access the 5G network through a NAN (e.g., gNB) of a RAN **204**. The NFs include an Authentication Server Function (AUSF) **206**, a Unified Data Management (UDM) **208**, an Access and Mobility management Function (AMF) **210**, a Policy Control Function (PCF) **212**, a Session Management Function (SMF) **214**, a User Plane Function (UPF) **216**, and a Charging Function (CHF) **218**.

(33) The interfaces N1 through N15 define communications and/or protocols between each NF as described in relevant standards. The UPF **216** is part of the user plane and the AMF **210**, SMF **214**, PCF **212**, AUSF **206**, and UDM **208** are part of the control plane. One or more UPFs can connect with one or more data networks (DNs) **220**. The UPF **216** can be deployed separately from control

plane functions. The NFs of the control plane are modularized such that they can be scaled independently. As shown, each NF service exposes its functionality in a Service Based Architecture (SBA) through a Service Based Interface (SBI) **221** that uses HTTP/2. The SBA can include a Network Exposure Function (NEF) **222**, a NF Repository Function (NRF) **224**, a Network Slice Selection Function (NSSF) **226**, and other functions such as a Security Edge Protection Proxy (SEPP) **230** or a Service Communication Proxy (SCP).

(34) The SBA can provide a complete service mesh with service discovery, load balancing, encryption, authentication, and authorization for interservice communications. The SBA employs a centralized discovery framework that leverages the NRF **224**, which maintains a record of available NF instances and supported services. The NRF **224** allows other NF instances to subscribe and be notified of registrations from NF instances of a given type. The NRF **224** supports service discovery by receipt of discovery requests from NF instances and, in response, details which NF instances support specific services.

(35) The NSSF **226** enables network slicing, which is a capability of 5G to bring a high degree of deployment flexibility and efficient resource utilization when deploying diverse network services and applications. A logical end-to-end (E2E) network slice has pre-determined capabilities, traffic characteristics, service-level agreements, and includes the virtualized resources required to service the needs of a Mobile Virtual Network Operator (MVNO) or group of subscribers, including a dedicated UPF, SMF, and PCF. The wireless device **202** is associated with one or more network slices, which all use the same AMF. A Single Network Slice Selection Assistance Information (S-NSSAI) function operates to identify a network slice. Slice selection is triggered by the AMF, which receives a wireless device registration request. In response, the AMF retrieves permitted network slices from the UDM **208** and then requests an appropriate network slice of the NSSF **226**.

(36) The UDM **208** introduces a User Data Convergence (UDC) that separates a User Data Repository (UDR) for storing and managing subscriber information. As such, the UDM **208** can employ the UDC under 3GPP TS 22.101 to support a layered architecture that separates user data from application logic. The UDM **208** can include a stateful message store to hold information in local memory or can be stateless and store information externally in a database of the UDR. The stored data can include profile data for subscribers and/or other data that can be used for authentication purposes. Given a large number of wireless devices that can connect to a 5G network, the UDM **208** can contain voluminous amounts of data that is accessed for authentication. Thus, the UDM **208** is analogous to a Home Subscriber Server (HSS), to provide authentication credentials while being employed by the AMF **210** and SMF **214** to retrieve subscriber data and context.

(37) The PCF **212** can connect with one or more application functions (AFs) **228**. The PCF **212** supports a unified policy framework within the 5G infrastructure for governing network behavior. The PCF **212** accesses the subscription information required to make policy decisions from the UDM **208**, and then provides the appropriate policy rules to the control plane functions so that they can enforce them. The SCP (not shown) provides a highly distributed multi-access edge compute cloud environment and a single point of entry for a cluster of network functions, once they have been successfully discovered by the NRF **224**. This allows the SCP to become the delegated discovery point in a datacenter, offloading the NRF **224** from distributed service meshes that make-up a network operator's infrastructure. Together with the NRF **224**, the SCP forms the hierarchical 5G service mesh.

(38) The AMF **210** receives requests and handles connection and mobility management while forwarding session management requirements over the N11 interface to the SMF **214**. The AMF **210** determines that the SMF **214** is best suited to handle the connection request by querying the NRF **224**. That interface and the N11 interface between the AMF **210** and the SMF **214** assigned by the NRF **224**, use the SBI **221**. During session establishment or modification, the SMF **214** also interacts with the PCF **212** over the N7 interface and the subscriber profile information stored

within the UDM **208**. Employing the SBI **221**, the PCF **212** provides the foundation of the policy framework which, along with the more typical QoS and charging rules, includes Network Slice selection, which is regulated by the NSSF **226**.

(39) The SEPP **230** facilitates secure interconnection between 5G networks. The SEPP **230** can route signaling messages between operator networks, serving as an interface for the wireless device **202** to operate on another operator's network as a roaming network.

(40) 4G Core Network Functions

(41) FIG. **3** is a block diagram that illustrates an architecture **300** including 4G core functions that can implement aspects of the present technology. A wireless device **302** can access the 4G network through a NAN (e.g., eNodeB **304**) of a RAN. The functions of the 4G architecture include a mobility management entity (MME) **306**, a serving gateway (SGW) **308**, a packet data network gateway (PDN gateway or PGW) **310**, a home subscriber service (HSS) **314**, a policy and charging rules functions (PCRF) **316**, and a diameter edge agent (DEA) **318**. These components can reside on nodes within a core packet-switched network operated by an LTE service provider, and their functionality can be split onto different physical nodes or merged onto shared-functionality nodes. Communications between the components can be enabled by interfaces such as virtual tunnels that are defined by relevant standards.

(42) The MME **306** provides mobility and session management to UEs. Operating as a network controller, the MME **306** can establish and maintain bearers as well as establish connection and security between the UE and the 4G core network. The HSS **314** stores data for customer profiles and creates authentication vectors for use by the MME **306**. The DEA **318** routes signaling messages between operator networks, serving as an interface for the wireless device **302** to operate on another operator's network as a roaming network.

(43) Each NAN **304** has a communication interface with the SGW **308**, which in turn has a communication interface with the PGW **310** that provides connectivity with an IP network **312** such as the Internet. The SGW **308** routes and forwards user data packets to or from the UE. The SGW **308** can furthermore facilitate handovers of the UE from the 4G network to another 4G network.

(44) The PGW **310** provides a UE with access to a PDN by assigning an Internet protocol (IP) address to the UE. In an LTE network, the PGW **310** can assign addresses based on both IP version 4 (IPv4) and IP version 6 (IPv6). The PGW **310** can further perform functions such as policy enforcement, packet filtering, and charging as packets are routed from the UE to the IP network **312** or from the IP network **312** to the UE. Quality of service information used by the PGW **310** can be supplied by the PCRF **316**, including charging rules, flow control rules, or traffic priority.

(45) Customized Roaming in Telecommunications Networks

(46) A telecommunications network has coverage areas, representing geographic areas in which the telecommunications network operates. For example, the coverage area of a network includes any geographic area within a specified distance from a NAN affiliated with the telecom provider, such that electronic devices within the specified distance from the NAN can communicate with the NAN. At times, it is beneficial for an electronic device that is registered to operate on a network maintained by a particular telecommunications provider to communicate with a NAN registered to a different telecommunications provider. For example, if the electronic device is operating in a geographic region that is outside the coverage area of its home network, the user of the electronic device may desire to access a roaming network from another telecommunications provider in order to enable the electronic device to send and receive data over the network. However, it can be expensive and burdensome for a telecommunications network to enable its electronic devices to operate on a roaming network.

(47) To balance the need for some electronic devices to access roaming networks against the cost and burden of maintaining the infrastructure to facilitate such roaming, a telecommunications network according to implementations herein facilitates selective roaming, in which some

electronic devices are allowed to operate on a roaming network while others are not. For example, a telecommunications network configures customer accounts for a subset of its customers to allow the customers in the subset to access roaming networks in particular geographic areas where the telecommunications network has limited or no coverage available.

(48) FIG. 4 illustrates an example network environment **400** in which selective roaming is enabled. In the environment **400**, electronic devices (including an electronic device **410A** associated with a customer A and an electronic device **410B** associated with a customer B) transmit and receive data over a network **440** by sending data to and receiving data from one or more network access nodes (NANs) **422, 432**.

(49) The electronic devices **410** are registered to a first telecommunications network and configured to operate on the first network. The NAN **422** is likewise associated with the first telecommunications network, and thus is part of a home public land mobile network (HPLMN) **420** that represents a combination of wireless communication services offered by the first telecommunications operator.

(50) The second NAN **432** is associated with a second telecommunications network. When the NAN **432** is used by the electronic devices registered to the first telecommunications network, the NAN **432** functions as part of a visiting public land mobile network (VPLMN) **430** that represents a combination of communication services offered by the second telecommunications operator on a roaming basis.

(51) In the network environment **400** according to implementations herein, the HPLMN **420** selectively permits electronic devices **410** to access the VPLMN **430** and associated roaming functionality provided by the second telecommunications network operator. For example, the electronic device **410A** associated with Customer A is permitted to access the VPLMN **430** when the device is located within the coverage area of the second telecommunications network, while the electronic device **410B** associated with Customer B is not permitted to access the roaming network.

(52) FIGS. 5-6 illustrate processes for selectively permitting access to a roaming network, according to some implementations. FIG. 5 illustrates an implementation of a selective roaming process **500** when the applicable telecommunications networks are 4G networks, while FIG. 6 illustrates an implementation of a selective roaming process **600** when the applicable telecommunications networks are 5G networks.

(53) As shown in FIG. 5, the process **500** includes interactions between devices implementing functionality of the HPLMN **420** (i.e., a first operator's network), devices implementing functionality of the VPLMN **430** (i.e., a second operator's network), and a roaming control system **520** associated with the HPLMN **420**. The process **500** begins when a UE device attaches to a VPLMN at step **501**. The attachment request is received at an MME **502** associated with the VPLMN **430**. Since the UE device is requesting to establish a connection as a visiting device rather than as a device registered to the network in which the MME **502** operates, the MME **502** passes the attachment request to a DEA **504** associated with the VPLMN **430**. The DEA **504** transmits a message, at step **503**, to a corresponding DEA **514** associated with the HPLMN **420** to notify the HPLMN of the UE's request to attach to the VPLMN **430**.

(54) The DEA **514** routes a message to a roaming control system **520** associated with the HPLMN **420** to enable the roaming control system **520** to determine whether to allow the UE to attach to the VPLMN **430**. In some implementations, the DEA **514** routes the message to the roaming control system **520** in step **505**, in which a copy of the message received from the DEA **504** is transmitted to the roaming control system **520**. In other implementations, the DEA **514** routes the message through the HPLMN's HSS **516** at step **507**, causing the HSS **516** in turn to pass either a copy of the received message or a message formatted according to the S6A protocol to the roaming control system **520**. In still other implementations, the DEA **514** directly routes an S6A message to the roaming control system **520**.

(55) A roaming controller **522** in the roaming control system **520** maintains mappings between UE

devices and roaming permissions for each device. When the roaming controller **522** receives the roaming request message routed through the DEA **514**, the controller **522** determines whether to grant or deny the roaming request based on the mappings. The roaming controller **522** queries account information associated with the UE to determine if the UE is authorized to access any roaming networks. For example, the roaming controller **522** queries, at step **511**, an account record **524** (maintained, for example, by a billing system or a provisioning system associated with the telecommunications network) to determine if the UE is authorized to access a roaming network. In some implementations, the account record **524** will include information about the particular roaming networks the UE is authorized to access (if any), such as identifiers of the geographic region(s) in which the authorized roaming networks operate, an identifier of the operator of the telecommunications network that is authorized for use as a roaming network, or other relevant information.

(56) The mappings maintained between UE devices and roaming positions can include one-to-one mappings between individual UE devices and individual geographic regions in which each device is permitted to roam, many-to-one mappings between groups of UE devices and individual geographic regions in which devices in the group are permitted to roam, one-to-many mappings between individual UE devices and groups of geographic regions in which each device is permitted to roam, or many-to-many mappings. The identifiers of UE devices used in the mappings can include identifiers of the UE devices themselves (e.g., an International Mobile Station Equipment Identity (IMEI), identifiers of subscriber identity modules (e.g., an International Mobile Subscriber Identity (IMSI), telephone numbers or ranges of telephone numbers, identifiers of users of the devices, or other such identifiers. These UE device identifiers can be mapped to geographic identifiers such as tracking area codes (TACs), which are identifiers defined within the first or second telecommunications network for each of a plurality of tracking areas covered by the respective networks. Other types of geographic identifiers that can be mapped to UE device identifiers in addition to or instead of TACs include, for example, zip codes, county identifiers, region identifiers, city identifiers, or state identifiers.

(57) The roaming controller **522** further queries the MME **502** (associated with the VPLMN **430**) for location information at step **513**. In some implementations, the roaming controller **522** uses an Insert Subscriber Data Request (IDR) command to retrieve the location information. The location information that is retrieved can include, for example, geographic coordinates of the UE at the time of the roaming request, geographic coordinates of the NAN to which the UE is requesting to attach, or an identifier of a geographic region in which the NAN is located.

(58) The roaming control system **520** applies decision logic **526** to determine whether to allow the UE roaming access to the VPLMN **430**. If the UE is authorized to access a roaming network in the particular geographic region where the access is requested, the decision logic **526** causes the roaming control system **520** to output an approval of the registration request at step **515**. If the UE is not authorized to access any roaming networks, or if the UE is requesting access to a roaming network in a non-approved geographic region, the decision logic **526** causes the roaming control system **520** to output a denial of the registration request at step **517**.

(59) The selective roaming process **600** shown in FIG. 6, as an example implementation of the process in a 5G network, is similar to the process **500** illustrated in FIG. 5 and similarly includes interactions between devices implementing functionality of the HPLMN **420**, devices implementing functionality of the VPLMN **430**, and a roaming control system **520** associated with the HPLMN **420**.

(60) Like the process **500**, the process **600** begins when a UE device attaches to a VPLMN at step **601**. The attachment request is received at an AMF **602** associated with the VPLMN **430**. The AMF **602** passes the attachment request to a SEPP **604** associated with the VPLMN **430**, which in turn transmits a message (step **603**) to a corresponding SEPP **614** associated with the HPLMN **420** to notify the HPLMN of the UE's request to attach to the VPLMN **430**.

(61) The SEPP **614** routes a message to the roaming control system **520** to enable the roaming control system **520** to determine whether to allow the UE to attach to the VPLMN **430**. As in the process **500**, different implementations of the SEPP **614** route the message to the roaming control system **520** in different ways, including the SEPP **614** routing a copy of the message directly to the roaming control system **520** (step **605**), the SEPP **614** routing the message through the HPLMN's UDM **616** (step **607**), or the SEPP **614** directly routing an S6A message to the roaming control system **520** (step **609**).

(62) The roaming controller **522** determines whether to grant or deny a roaming request based on (1) whether the UE is authorized to access roaming networks, and (2) whether the UE is authorized to use a roaming network in the particular geographic region in which the UE is requesting access. The roaming controller **522** can query the account record **524** at step **611** for roaming authorization and the AMF **602** at step **613** for location information, which are input to the decision logic **526** to determine whether to allow the UE roaming access to the VPLMN **430**. If the UE is authorized to access a roaming network in the particular geographic region where the access is requested, the decision logic **526** causes the roaming control system **520** to output an approval of the registration request at step **615**. If the UE is not authorized to access any roaming networks, or if the UE is requesting access to a roaming network in a non-approved geographic region, the decision logic **526** causes the roaming control system **520** to output a denial of the registration request at step **617**.

(63) Disabling Roaming in Overlapping Coverage Area

(64) When a telecommunications network allows a UE device to attach to a roaming network, the telecommunications network needs a mechanism to bring the UE device back to the home network when roaming is no longer needed. As described in the following, the UE device can be forced to reattach to the home network when the UE device is located in an overlapping coverage area, which is a geographic area in which coverage is available from both the home network and the roaming network. However, determining the UE device is in an overlapping coverage area represents merely an example policy under which the device is brought back to the home network. The telecommunications network can apply other policies, in addition to or instead of, the overlapping coverage policy. For example, the telecommunications network may allow the UE device to remain attached to a roaming network until the signal strength of the home network is greater than the signal strength available from the roaming network. In another example, the telecommunications network determines whether to allow the UE device to remain attached to the roaming network based on services being utilized by the UE. If, for example, the roaming network is a 5G network while the home network is a 4G network, the telecommunications network may permit the UE device to continue using the roaming network when the UE device is accessing a data-heavy application that requires low latency.

(65) FIG. 7 illustrates example coverage areas of the HPLMN **420** (a “home network coverage area **720**”) and the VPLMN **430** (a “visiting network coverage area **730**”). In some areas, the home network coverage area **720** and the visiting network coverage area **730** overlap, creating an overlapping coverage area **740**. Each of the coverage areas **720**, **730**, **740** can include one or more tracking areas, each represented by a respective tracking area code. When a UE has been allowed access to the VPLMN **430** (e.g., because the UE is located within the visiting network coverage area **730**), the HPLMN **420** monitors for movement of the UE into the overlapping coverage area **740**. When the UE is determined to have moved into the overlapping coverage area **740**, the HPLMN **420** initiates a process to return the UE to the home network rather than continuing to allow the UE to access the roaming network.

(66) FIG. 8 illustrates a first example process **800** to disconnect a UE from a roaming network when the UE is located in an overlapping coverage area between an HPLMN and a VPLMN. In general, the process in FIG. 8 includes steps that are similar to steps of the processes for connecting a UE to a roaming network described in FIGS. 5 and 6, but instead detects that a UE has moved from a coverage area of the roaming network to an overlapping coverage area between the roaming

network and the home network.

(67) As shown in FIG. 8, the process **800** begins when a UE issues a tracking area update request **801** that causes the UE to detach from an old MME **502** and attach to a new MME **802**. The new MME **802** passes the attachment request to the DEA **504** at step **803**, which in turn transmits a message (at step **805**) to the DEA **514** associated with the HPLMN **420** to notify the HPLMN of the UE's request to attach to the new MME **802**. The DEA **514** routes a message to the roaming control system **520** to enable the roaming control system to determine whether to allow the UE to attach to the new MME **802**. As in the processes described in FIGS. 5 and 6, the message can be routed between the DEA **514** and the roaming control system **520** in various ways, including directly transmitting a copy of the attachment request from the DEA **514**, causing the HSS **516** to transmit a copy of the attachment request, or sending an S6a message via the DEA **514** or HSS **516** to the roaming control system **520**.

(68) The roaming controller **522** queries the new MME **802** for location information associated with the UE at step **813**. The location information can include, for example, a tracking area code identifying a tracking area in which the UE is located, global positioning coordinates of the UE, an identity of a NAN to which the UE is communicatively coupled, and/or other location identifiers that can be used by the roaming controller **522** to determine the location of the UE.

(69) The decision logic **526** is applied by the roaming control system **520** to determine if the UE has moved into the overlapping coverage area between the VPLMN and the HPLMN. In various implementations, the roaming control system **520** determines the UE has moved into the overlapping coverage area **740** by determining a tracking area code used by the UE is associated with a tracking area in the overlapping coverage area **740**. Alternatively, the roaming control system **520** determines based on global positioning coordinates or other location identifiers that the UE is located within an explicit boundary region defining the overlapping coverage area **740**, within a boundary of the coverage area of the HPLMN, within a communicable distance to a NAN associated with the HPLMN (such as the NAN **422**), etc. For example, the roaming controller **522** accesses a data record that defines a location of a geographic boundary around the overlapping coverage area or the coverage area of the HPLMN. The decision logic **526** causes the roaming control system **520** to determine the UE has moved into the overlapping coverage area if the geographic location of the UE is inside an area defined by the boundary or less than a threshold distance outside the boundary.

(70) If the UE is determined to be located within a coverage area of the HPLMN (including in the overlapping coverage area), the roaming control system **520** denies the registration of the UE to the new MME **802** at step **817**. Once the UE has been denied registration to the new MME **802**, a registration to the HPLMN can be initiated to allow the UE to continue operating on its home network.

(71) FIG. 9 illustrates a second example process **900** to disconnect a UE from a roaming network. The process **900** can be performed, for example, when the UE does not issue a tracking area update (e.g., because the UE can remain attached to the original MME **502** when it moves into the overlapping coverage area). However, the process **900** can also be performed when the UE attaches to a new MME.

(72) As shown in FIG. 9, an SGW **902** associated with the VPLMN **430** transmits a modify bearer request (MBR) **903** to a PGW **904** associated with the HPLMN **420**. The MBR **903** is triggered, for example, when the MME **502** detects the UE has moved.

(73) The PGW **904** interfaces with a data probe **906**, which is configured for example as a proxy server that monitors MBRs received from the VPLMN **430**. The data probe **906** communicates a feed of information to the roaming controller **522** at step **905**, including any received MBRs.

(74) The roaming controller **522** decodes user location information (ULI) from the feed of information received from the data probe **906**. The ULI includes an identifier of the UE and the location of the UE (e.g., as a tracking area code). If the location of the UE falls within the

overlapping area between the VPLMN **430** and the HPLMN **420** (as determined, for example, according to any of the processes described above with respect to FIG. **8**), the roaming controller **522** outputs a command to block the UE's access to the roaming network at step **909**. For example, the roaming controller **522** sends the MME **502** a cancel-location-request (CLR) message to withdraw the UE's subscription to the roaming MME **502**.

(75) FIG. **10** illustrates a third example process **1000** to disconnect a UE from a roaming network. Like the process **900** described with respect to FIG. **9**, the process **1000** can be performed regardless of whether the UE issues a tracking area update. In general, the process **1000** configures the roaming controller **522** as a network proxy device that receives data about location changes of the UE and causes the UE's subscription to the roaming network to be canceled when the UE enters the overlapping area between the VPLMN **430** and the HPLMN **420**.

(76) At step **1001**, the SGW **902** associated with the VPLMN **430** issues a query for an access point name (APN) for a gateway that is usable by the UE to access a computer network other than the VPLMN **430**. The APN query **1001**, which can be issued by the SGW **902** at any point while the UE is attached to the VPLMN **430**, is intercepted by a roaming domain name server (DNS) **1002** associated with the HPLMN **420**. The roaming DNS **1002** responds to the APN query **1001** with an IP address of the roaming controller **522**, to configure the roaming controller **522** as a network proxy. In particular, some implementations of the roaming controller **522** function as a general packet radio service (GPRS) tunnelling protocol (GTP) controller that interfaces with the SGW **902** and PGW **904**. In the roaming controller's function as a GTP controller, the SGW **902** transmits GTP control (GTP-C) messages to the roaming controller **522** at step **1005** and GTP user data tunneling (GTP-U) messages to the PGW **904** at step **1007**. Likewise, the roaming controller **522** sends GTP-C messages to the PGW **904** at step **1009**. When the VPLMN is a 5G network, the roaming controller **522** can similarly interface with an SMF and UPF within the roaming telecommunications network.

(77) When the UE changes location, a command to modify the UE's session is transmitted from the SGW **902** to the roaming controller **522** as the GTP-C command **1005**. For example, a modify bearer request is sent via the GTP-C plane to the roaming controller **522**. Upon receiving the request, the roaming controller **522** queries the account record **524** at step **1011** to determine if the new location of the UE is within the overlapping coverage area between the VPLMN and the HPLMN. If so, the roaming controller **522** is a command, at step **1013**, to block the UE's access to the roaming network. For example, the roaming controller **522** sends the MME **502** a cancel-location-request (CLR) message to withdraw the UE's subscription to the roaming MME **502**.

(78) Although the processes **800**, **900**, and **1000** are described with respect to components and functions of a 4G LTE network, similar processes can be implemented using analogous components and functions of a 5G NR network, or with a mix of components and functions of a 4G network and a 5G network.

(79) Computer System

(80) FIG. **11** is a block diagram that illustrates an example of a computer system **1100** in which at least some operations described herein can be implemented. As shown, the computer system **1100** can include: one or more processors **1102**, main memory **1106**, non-volatile memory **1110**, a network interface device **1112**, video display device **1118**, an input/output device **1120**, a control device **1122** (e.g., keyboard and pointing device), a drive unit **1124** that includes a storage medium **1126**, and a signal generation device **1130** that are communicatively connected to a bus **1116**. The bus **1116** represents one or more physical buses and/or point-to-point connections that are connected by appropriate bridges, adapters, or controllers. Various common components (e.g., cache memory) are omitted from FIG. **11** for brevity. Instead, the computer system **1100** is intended to illustrate a hardware device on which components illustrated or described relative to the examples of the figures and any other components described in this specification can be implemented.

(81) The computer system **1100** can take any suitable physical form. For example, the computing system **1100** can share a similar architecture as that of a server computer, personal computer (PC), tablet computer, mobile telephone, game console, music player, wearable electronic device, network-connected (“smart”) device (e.g., a television or home assistant device), AR/VR systems (e.g., head-mounted display), or any electronic device capable of executing a set of instructions that specify action(s) to be taken by the computing system **1100**. In some implementation, the computer system **1100** can be an embedded computer system, a system-on-chip (SOC), a single-board computer system (SBC) or a distributed system such as a mesh of computer systems or include one or more cloud components in one or more networks. Where appropriate, one or more computer systems **1100** can perform operations in real-time, near real-time, or in batch mode.

(82) The network interface device **1112** enables the computing system **1100** to mediate data in a network **1114** with an entity that is external to the computing system **1100** through any communication protocol supported by the computing system **1100** and the external entity. Examples of the network interface device **1112** include a network adaptor card, a wireless network interface card, a router, an access point, a wireless router, a switch, a multilayer switch, a protocol converter, a gateway, a bridge, bridge router, a hub, a digital media receiver, and/or a repeater, as well as all wireless elements noted herein.

(83) The memory (e.g., main memory **1106**, non-volatile memory **1110**, machine-readable medium **1126**) can be local, remote, or distributed. Although shown as a single medium, the machine-readable medium **1126** can include multiple media (e.g., a centralized/distributed database and/or associated caches and servers) that store one or more sets of instructions **1128**. The machine-readable (storage) medium **1126** can include any medium that is capable of storing, encoding, or carrying a set of instructions for execution by the computing system **1100**. The machine-readable medium **1126** can be non-transitory or comprise a non-transitory device. In this context, a non-transitory storage medium can include a device that is tangible, meaning that the device has a concrete physical form, although the device can change its physical state. Thus, for example, non-transitory refers to a device remaining tangible despite this change in state.

(84) Although implementations have been described in the context of fully functioning computing devices, the various examples are capable of being distributed as a program product in a variety of forms. Examples of machine-readable storage media, machine-readable media, or computer-readable media include recordable-type media such as volatile and non-volatile memory devices **1110**, removable flash memory, hard disk drives, optical disks, and transmission-type media such as digital and analog communication links.

(85) In general, the routines executed to implement examples herein can be implemented as part of an operating system or a specific application, component, program, object, module, or sequence of instructions (collectively referred to as “computer programs”). The computer programs typically comprise one or more instructions (e.g., instructions **1104**, **1108**, **1128**) set at various times in various memory and storage devices in computing device(s). When read and executed by the processor **1102**, the instruction(s) cause the computing system **1100** to perform operations to execute elements involving the various aspects of the disclosure.

REMARKS

(86) The terms “example”, “embodiment” and “implementation” are used interchangeably. For example, reference to “one example” or “an example” in the disclosure can be, but not necessarily are, references to the same implementation; and, such references mean at least one of the implementations. The appearances of the phrase “in one example” are not necessarily all referring to the same example, nor are separate or alternative examples mutually exclusive of other examples. A feature, structure, or characteristic described in connection with an example can be included in another example of the disclosure. Moreover, various features are described which can be exhibited by some examples and not by others. Similarly, various requirements are described which can be requirements for some examples but no other examples.

(87) The terminology used herein should be interpreted in its broadest reasonable manner, even though it is being used in conjunction with certain specific examples of the invention. The terms used in the disclosure generally have their ordinary meanings in the relevant technical art, within the context of the disclosure, and in the specific context where each term is used. A recital of alternative language or synonyms does not exclude the use of other synonyms. Special significance should not be placed upon whether or not a term is elaborated or discussed herein. The use of highlighting has no influence on the scope and meaning of a term. Further, it will be appreciated that the same thing can be said in more than one way.

(88) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” As used herein, the terms “connected,” “coupled,” or any variant thereof means any connection or coupling, either direct or indirect, between two or more elements; the coupling or connection between the elements can be physical, logical, or a combination thereof. Additionally, the words “herein,” “above,” “below,” and words of similar import can refer to this application as a whole and not to any particular portions of this application. Where context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. The word “or” in reference to a list of two or more items covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list. The term “module” refers broadly to software components, firmware components, and/or hardware components.

(89) While specific examples of technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize. For example, while processes or blocks are presented in a given order, alternative implementations can perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified to provide alternative or sub-combinations. Each of these processes or blocks can be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in series, these processes or blocks can instead be performed or implemented in parallel, or can be performed at different times. Further, any specific numbers noted herein are only examples such that alternative implementations can employ differing values or ranges.

(90) Details of the disclosed implementations can vary considerably in specific implementations while still being encompassed by the disclosed teachings. As noted above, particular terminology used when describing features or aspects of the invention should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features, or aspects of the invention with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the invention to the specific examples disclosed herein, unless the above Detailed Description explicitly defines such terms. Accordingly, the actual scope of the invention encompasses not only the disclosed examples, but also all equivalent ways of practicing or implementing the invention under the claims. Some alternative implementations can include additional elements to those implementations described above or include fewer elements.

(91) This patent application is related to U.S. patent application Ser. No. 17/897,075, filed Aug. 26, 2022, now U.S. Pat. No. 12,225,623, which is incorporated herein by reference in its entirety. In addition, any patents and applications and other references noted above, and any that may be listed in accompanying filing papers, are incorporated herein by reference in their entireties, except for any subject matter disclaimers or disavowals, and except to the extent that the incorporated material is inconsistent with the express disclosure herein, in which case the language in this disclosure controls. Aspects of the invention can be modified to employ the systems, functions, and

concepts of the various references described above to provide yet further implementations of the invention.

(92) To reduce the number of claims, certain implementations are presented below in certain claim forms, but the applicant contemplates various aspects of an invention in other forms. For example, aspects of a claim can be recited in a means-plus-function form or in other forms, such as being embodied in a computer-readable medium. A claim intended to be interpreted as a mean-plus-function claim will use the words “means for.” However, the use of the term “for” in any other context is not intended to invoke a similar interpretation. The applicant reserves the right to pursue such additional claim forms in either this application or in a continuing application.

Claims

1. A system comprising: at least one hardware processor; and at least one non-transitory memory storing instructions, which, when executed by the at least one hardware processor, cause the system to: monitor a Modify Bearer Request (MBR) generated by a first cellular network for a wireless device, wherein the first cellular network is operating as a roaming network for the wireless device, wherein the wireless device is registered to operate on a second cellular network, and wherein the first cellular network and second cellular network each have a respective coverage area; without receiving a tracking area update from the wireless device, decode the MBR to determine user location information associated with the wireless device; and in response to the user location information indicating the wireless device is located in an overlapping area between the coverage area of the first cellular network and the coverage area of the second cellular network, issue a cancel location request to cause the wireless device to disconnect from the first cellular network.
2. The system of claim 1, wherein monitoring the MBR comprises: receiving the MBR via a data probe coupled to a gateway associated with the second cellular network.
3. The system of claim 1, wherein the instructions when executed further cause the system to: interface with a serving gateway (SGW) associated with the first cellular network to receive general packet radio service (GPRS) tunnelling protocol control plane (GTP-C) messages over the interface; wherein the MBR is transmitted via one of the GTP-C messages.
4. The system of claim 1, wherein the instructions when executed further cause the system to: interface with a device implementing a session management function (SMF) associated with the first cellular network to receive general packet radio service (GPRS) tunnelling protocol control plane (GTP-C) messages over the interface; wherein the MBR is transmitted via one of the GTP-C messages.
5. The system of claim 1, wherein the user location information includes a tracking area code indicating a tracking area in which the wireless device is located, and wherein the instructions further cause the system to: access a data record that includes one or more tracking areas that are within the overlapping area; and determine the wireless device is located in the overlapping area if the tracking area code corresponds to one of the tracking areas within the overlapping coverage area.
6. The system of claim 1, wherein the user location information includes a tracking area code indicating a tracking area in which the wireless device is located, and wherein the instructions further cause the system to: access a data record that includes one or more tracking areas that are within the coverage area of the second cellular network; and determine the wireless device is located in the overlapping area if the tracking area code corresponds to one of the tracking areas within the coverage area of the second cellular network.
7. The system of claim 1, wherein the instructions further cause the system to: access a data record that defines a geographic boundary around the coverage area of the second cellular network; and determine the wireless device is located in the overlapping area if the user location information indicates the wireless device is inside the geographic boundary defined in the data record.

8. A non-transitory computer-readable storage medium comprising instructions recorded thereon, wherein the instructions when executed by at least one data processor of a system, cause the system to: monitor changes in location of a user equipment (UE) device while the UE device is subscribed to a roaming telecommunications network, wherein the UE device is registered to operate on a home telecommunications network different from the roaming telecommunications network, and wherein each of the roaming telecommunications network and the home telecommunications network have a coverage area; and without receiving a tracking area update from the UE device, in response to detecting the UE device has moved from a first geographic location that is within the coverage area of the roaming telecommunications network and not the home telecommunications network, to a second geographic location that is within both the coverage area of the roaming telecommunications network and the coverage area of the home telecommunications network, generate a command to cancel the subscription of the UE device to the roaming telecommunications network.

9. The non-transitory computer-readable storage medium of claim 8, wherein monitoring changes in the location of the UE device comprises: receiving a Modify Bearer Request (MBR) via a data probe coupled to a gateway associated with the roaming telecommunications network; and decoding the MBR to determine a new location of the UE device.

10. The non-transitory computer-readable storage medium of claim 8, wherein monitoring changes in the location of the UE device comprises: interfacing with a serving gateway (SGW) associated with the roaming telecommunications network to receive general packet radio service (GPRS) tunnelling protocol control plane (GTP-C) messages over the interface; and decoding a message of the received GTP-C messages to determine a new location of the UE device.

11. The non-transitory computer-readable storage medium of claim 8, wherein the instructions when executed further cause the system to: interface with a device implementing a session management function (SMF) associated with the roaming telecommunications network to receive general packet radio service (GPRS) tunnelling protocol control plane (GTP-C) messages over the interface; and decode a message of the received GTP-C messages to determine a new location of the UE device.

12. The non-transitory computer-readable storage medium of claim 8, wherein monitoring changes in the location of the UE device comprises receiving a tracking area code indicating a tracking area in which the UE device is located, and wherein the instructions further cause the system to: access a data record that includes one or more tracking areas that are within both the coverage area of the roaming telecommunications network and the coverage area of the home telecommunications network; and determine the UE device is located in an overlapping area if the tracking area code corresponds to one of the tracking areas within both the coverage area of the roaming telecommunications network and the coverage area of the home telecommunications network.

13. The non-transitory computer-readable storage medium of claim 8, wherein generating the command to cancel the subscription of the UE device to the roaming telecommunications network comprises generating a cancel location request (CLR) message.

14. A non-transitory, computer-readable storage medium comprising instructions recorded thereon, wherein the instructions when executed by at least one data processor of a system, cause the system to: monitor changes in location of a user equipment (UE) device while the UE device is subscribed to a first telecommunications network; interface with a device implementing a session management function (SMF) associated with the first telecommunications network to receive general packet radio service (GPRS) tunnelling protocol control plane (GTP-C) messages over the interface; decode a message of the received GTP-C messages to determine a new location of the UE device; communicate a request to update a location of the UE device, in response to determining the new location of the UE device, to a second telecommunications network that operates as a home network for the UE device; and if a response to the request to update the location of the UE device indicates the UE device has moved into an area of overlapping coverage area between the first

telecommunications network and the second telecommunications network, cancel the subscription of the UE device to the first telecommunications network.

15. The non-transitory, computer-readable storage medium of claim 14, wherein the instructions further cause the system to: receive, in a first telecommunications network, a tracking area update for a user equipment (UE) device subscribed to operate on the first telecommunications network as a roaming network, wherein the tracking area update is received from a mobility management entity (MME) associated with the first telecommunications network.

16. The non-transitory, computer-readable storage medium of claim 15, wherein the tracking area update is received from a device implementing an access and mobility management function (AMF) associated with the first telecommunications network.

17. The non-transitory, computer-readable storage medium of claim 14, wherein the request to update the location of the UE device is communicated to a diameter edge agent (DEA) associated with the second telecommunications network.

18. The computer-readable storage medium of claim 14, wherein the request to update the location of the UE device is communicated to a security edge protection proxy (SEPP) associated with the second telecommunications network.

19. The non-transitory, computer-readable storage medium of claim 14, wherein the response to the request to update the location of the UE device comprises a cancel location request (CLR) message.
